Preface

This project investigates the field of electric vehicle (EV) sales forecasting and analysis, which is critical for understanding market trends and making informed decisions in the growing EV industry. With the increasing adoption of electric vehicles and the need for precise sales predictions, developing accurate forecasting models is essential.

In this project, we focus on automating the data collection process from the "Vahan Parivahan" government website and creating a seamless data processing pipeline for structured storage in a PostgreSQL database. The collected data is then analyzed using time series forecasting models, specifically LSTM (Long Short-Term Memory) and ARIMA (Autoregressive Integrated Moving Average), to predict future sales trends.

Our methodology encompasses data collection, preprocessing, model building, and evaluation. The LSTM and ARIMA models are trained and compared to determine the best-performing model for EV sales forecasting. The project also involves creating an interactive Power BI dashboard to visualize and analyze the forecasted data, providing stakeholders with actionable insights.

This project aims to enhance the understanding of EV sales trends and contribute to the industry's growth by providing reliable sales predictions. The findings and insights from this project can support stakeholders, policymakers, industry analysts, and enthusiasts in making data-driven decisions and promoting the adoption of electric vehicles.